

FARZAN K MALEKABADI

Email: karimima@usc.edu ◇ Website
Permanent Resident (U.S. Green Card Holder)

EDUCATION

Ph.D., Social Psychology (University of Southern California), 2021 – 2025
M.S. & B.S., Engineering

SKILLS

Programming & Data: Python, R, SQL, Databricks, Git

Statistical Modeling: Causal Inference, Bayesian Modeling, Multilevel Modeling, Time-Series Analysis, EFA/CFA/SEM, Psychometrics (CTT, IRT)

AI Evaluation & NLP: LLM Evaluation, Agentic Systems, Prompt Evaluation, NLP, Classification Modeling, Human-in-the-Loop Evaluation

Research Methods: Experimental Design, Survey Research, A/B Testing, UX Research, Mixed-Methods Research, Inter-Rater Reliability

RESEARCH EXPERIENCE

Google — Quantitative User Research Intern

May 2025 — August 2025

- **User Segmentation and Journey Validation**

Led a Critical User Journey (CUJ) validation and segmentation study for a key privacy feature. Applied Latent Class Analysis and multilevel A/B testing to uncover user personas and quantify adoption likelihood and trust sentiment, informing product strategy for privacy communications.

- **End-to-End Research for Agentic AI Product**

Designed and executed end-to-end quantitative research for a new agentic AI product, from scoping and alignment sessions to readout, collaborating with PMs, designers, and engineers to assess user trust and data-sharing behavior. Delivered insights in under one month that informed PM decisions on the welcome screen design.

Edmentum — Research Scientist

June 2024 — Present

- **AI Chatbot Evaluation & UX Research**

Designed evaluation protocols for an AI-powered educational chatbot, conducting qualitative interviews with middle school students and quantitative UX research. Led SME calibration sessions, calculated inter-rater reliability, and developed standardized rubrics and evaluation scenarios to assess response quality and user experience.

- **AI Content Evaluation & Agentic Generation**

Built automated Python pipelines to evaluate AI-generated educational content, benchmarking GPT models on 12K item-skill pairs using precision, recall, and F1 metrics. Developed multi-agent generation and review systems with feedback-driven quality controls and simulation-based validation to improve the reliability and scalability of assessment content pipelines.

- **Diagnostic Assessment Analytics**

Led end-to-end quantitative research on student testing behavior to optimize diagnostic assessment length. Built ELT pipelines in Databricks to scope, deduplicate, and analyze billions of item-level assessment data points, applying statistical analysis in R and creating dashboards and automated reporting workflows that informed product decisions and addressed customer feedback.

- **Psychometric Modeling & Item Quality Evaluation**

Built data pipelines to estimate Classical Test Theory (CTT) and Item Response Theory (IRT) models on operational assessment data. Generated empirical evidence on test reliability and item difficulty to guide content expansion and assessment strategy.

University of Southern California — Research Fellow

May 2021 — Jan 2026

- **Quantitative and Computational Social Psychology**

Conduct empirical and theoretical research on moral cognition, cultural evolution, and human–AI reasoning. Designed and analyzed large-scale cross-cultural surveys and panel datasets (over 300,000 participants) using causal inference, Bayesian hierarchical modeling, and multilevel regression.

- **Longitudinal, Time-Series, and Geospatial Modeling**

Integrated historical, ecological, and cultural indicators to study the dynamics of moral and cultural change across time and regions using panel, time-series, and spatial analyses.

- **Natural Language Processing and Computational Modeling**

Developed multilingual moral language corpora and AI reasoning benchmarks. Applied NLP, moral sentiment classification, and LLM evaluation frameworks to investigate cross-cultural variation in moral reasoning.

- **Theory Building and Measurement**

Advanced theoretical models of moral and cultural psychology. Designed and validated psychometric instruments through EFA, CFA, and SEM.

- **Undergraduate Statistics Instruction**

Led lab sessions on statistical analysis using SPSS and R, covering descriptive and inferential statistics including t-tests, ANOVA, correlation, regression, and chi-square tests. Guided students through applied data analysis, visualization, and interpretation emphasizing reproducibility and clear statistical reasoning.

PUBLICATION AND PREPRINTS

1. **Karimi-Malekabadi, F.**, Abdurahman, S., Sourati, Z., Trager, J., & Dehghani, M. (2026). Theory Trace Card: Theory-Driven Socio-Cognitive Evaluation of LLMs. *arXiv*. <https://arxiv.org/abs/2601.01878>
2. **Karimi-Malekabadi, F.**, Razavi, P., & Powers, S. (2025). Scaling item-to-standard alignment with large language models: Accuracy, limits, and solutions. *PsyArXiv*. https://doi.org/10.31234/osf.io/cyqs7_v1
3. Abdurahman, S., **Karimi-Malekabadi, F.**, Yu, C., Kteily, N. S., & Dehghani, M. (2025). Realistic threat perception drives intergroup conflict: A causal, dynamic analysis using generative-agent simulations. *arXiv*. <https://arxiv.org/abs/2512.17066>
4. Yu, C., Yi, B., **Karimi-Malekabadi, F.**, Abdurahman, S., Ye, J., Narayanan, S., ... & Dehghani, M. (2026). Tracing Moral Foundations in Large Language Models. *arXiv*. <https://arxiv.org/abs/2601.05437>
5. Trager, J., **Karimi-Malekabadi, F.**, Abdurahman, S., Daryani, Y., & Dehghani, M. (2026). The Power of Social Media Bios: How Common Profile Content Unintentionally Signals Political Ideology and Shapes Prejudice. *OSF Preprints*. https://doi.org/10.31234/osf.io/8yzh6_v1
6. Ziabari, A. S., Ghazizadeh, N., Sourati, Z., **Karimi-Malekabadi, F.**, Piray, P., & Dehghani, M. (2025). Reasoning on a Spectrum: Aligning LLMs to System 1 and System 2 Thinking.

7. Sourati, Z., **Karimi-Malekabadi, F.**, Ozcan, M., McDaniel, C., Ziabari, A., Wen, N., Tak, A., ... & Dehghani, M. (in press). The Shrinking Landscape of Linguistic Diversity in the Age of Large Language Models. *Nature Human Behaviour*. <https://arxiv.org/abs/2502.11266>
8. Abdurahman, S., Atari, M., **Karimi-Malekabadi, F.**, Xue, M. J., Trager, J., Park, P. S., Golazizian, P., Omrani, A., & Dehghani, M. (2024). Perils and opportunities in using large language models in psychological research. *PNAS Nexus*, 3(7), e245. <https://doi.org/10.31234/osf.io/d695y>
9. **Karimi-Malekabadi, F.**, Oyserman, D. (2025). Costly Morality Theory of Honor: An Evolutionary, Culture-as-Situated Cognition Perspective. *Personality and Social Psychology Review*. https://doi.org/10.31219/osf.io/n6v74_v1
10. **Karimi-Malekabadi, F.**, Sachdeva, S., & Dehghani, M. (2025). A value-based topography of climate change beliefs and behaviors. *PNAS Nexus*, 4(2), pgae590. <https://doi.org/10.1093/pnasnexus/pgae590>
11. **Karimi-Malekabadi, F.**, Ziabari, A. S., & Dehghani, M. (2025). Revisiting National Nostalgia: Make America Moral Again! *PsyArXiv*. https://doi.org/10.31234/osf.io/xvdep_v1
12. **Karimi-Malekabadi, F.**, Oyserman, D., Ziabari, A., & Omrani, A. (2024). What Doesn't Kill Us: Harsh Ecologies Beget Difficulty-as-Improvement Norms. *Preprint*. <https://doi.org/10.21203/rs.3.rs-4110737/v1>
13. Trager, J., Vargas, F., Alves, D., Guida, M., Ngueajio, M. K., Agrawal, A., Plaza-del-Arco, F., Daryanai, Y., & **Karimi-Malekabadi, F.** (2025). MFTCXplain: A Multilingual Benchmark Dataset for Evaluating the Moral Reasoning of LLMs through Hate Speech Multi-hop Explanation. *Findings of the Association for Computational Linguistics: EMNLP 2025*.
14. Trager, J., Ziabari, A. S., Davani, A. M., Golazizian, P., **Karimi-Malekabadi, F.**, Omrani, A., ... & Dehghani, M. (2026). The Moral Foundations Reddit Corpus. In *Proceedings of the 15th Conference on Language Resources and Evaluation (LREC 2026)*. <https://arxiv.org/abs/2208.05545>
15. Trager, J., **Karimi-Malekabadi, F.**, Abdurahman, S., & Dehghani, M. (in press). Hate is justified when values are threatened: Evidence from ideologically threatening tweets and real world events. *PLOS ONE*. <https://osf.io/download/5muq7>
16. **Karimi-Malekabadi, F.**, & Falahatpishe Baboli, M. (2022). Qeirat Values and Victim Blaming in Iran: The Mediating Effect of Culture-Specific Gender Roles. *Journal of Interpersonal Violence*. <https://doi.org/10.1177/08862605221101184>
17. Reimer, N. K., Atari, M., **Karimi-Malekabadi, F.**, Trager, J., Kennedy, B., Graham, J., & Dehghani, M. (2022). Moral values predict county-level COVID-19 vaccination rates in the United States. *American Psychologist*, 77(6), 743. <https://doi.org/10.1037/amp0001020>
18. Bago, B., Kovacs, M., Protzko, J., Nagy, T., Kekecs, Z., Palfi, B., ... **Karimi-Malekabadi, F.** & Matibag, C. J. (2022). Situational factors shape moral judgements in the trolley dilemma in Eastern, Southern and Western countries. *Nature Human Behaviour*. <https://doi.org/10.1038/s41562-022-01319-5>
19. Atari, M., Reimer, N. K., Graham, J., Hoover, J., Kennedy, B., Davani, A. M., **Karimi-Malekabadi, F.**, Birjandi, S., & Dehghani, M. (2022). Pathogens are linked to human moral systems across time and space. *Current Research in Ecological and Social Psychology*, 3, 100060. <https://doi.org/10.1016/j.cresp.2022.100060>

20. **Karimi-Malekabadi, F.**, Esmailinasab, M. (2019). Religiosity, intrasexual rivalry, and mate retention behaviors in Iran. *Personality and Individual Differences*, 149, 135-140. <https://doi.org/10.1016/j.paid.2019.05.059>
21. **Karimi-Malekabadi, F.**, Atari, M., & Dehghani, M. (2024). Mobile sensing applications in moral psychology. In M. R. Mehl, M. Eid, C. Wrzus, G. M. Harari, & U. W. Ebner-Priemer (Eds.), *Mobile sensing in psychology: Methods and applications*. Guilford Press.
22. **Karimi-Malekabadi, F.**, Ghanbarian, E., Afhami, R., & Chegeni, R. (2019). Theory-driven assessment of intrasexual rivalry. *Evolutionary Psychological Science*, 5(3), 286-293. <https://doi.org/10.1007/s40806-018-00185-7>